Chapter 6: Samples and Observational Studies

The Practice of Statistics in the Life Sciences
by Brigitte Baldi and David S. Moore

Instructor: Ali Raisolsadat

Part III: Foundations of Statistical Inference

Why This Chapter Matters

Statistical methods are useful only when the data have been collected in a way that matches the scientific question.

Before a mean, proportion, confidence interval or test is interpreted, the study design should make clear

- who or what is being studied
- what variables are being measured
- how the observations are obtained

The quality of a statistical conclusion depends on the quality of the data-generation process.

Gathering Data and Study Design

- Many scientific questions concern populations that are too large to study completely.
  1. A health researcher may want to estimate a proportion for adults in a country.
  2. A wildlife biologist may want to learn about fish in a river system.
  3. A laboratory may want to understand a biological process from a manageable set of specimens.
- In each setting, the main issue is not simply how much data are collected. The main issue is whether the data were collected through a design that supports the intended conclusion.
- A numerical method cannot repair a study design that systematically excludes important groups, measures the wrong variable or mixes the effect of an explanatory variable with other factors.
- Statistical reasoning therefore begins with a clear description of how the data were generated.

Observational Studies versus Experiments

Suppose researchers want to know whether a fertilizer increases seedling growth.

Observational Study: Observe Existing Conditions

Researchers compare seedlings that growers already fertilize with seedlings they do not fertilize. The researchers record conditions and growth without assigning fertilizer use.

Experiment: Assign Treatments

Researchers assign seedlings to receive fertilizer or water alone then compare their growth over four weeks.

In the observational study, the groups might also differ in soil, sunlight, or watering. These differences could help explain any observed difference in growth.

Random Sampling versus Random Assignment

Suppose a greenhouse contains 1000 seedlings.

Random Sampling: Which Seedlings Enter the Study?

Researchers randomly select 60 seedlings from the greenhouse.
Random sampling helps the sample represent the greenhouse population, supporting generalization to that population.

Random Assignment: Which Treatment Does Each Receive?

Researchers randomly assign the selected seedlings: 30 receive fertilizer and 30 receive water alone.
Random assignment makes the treatment groups comparable on average, supporting a causal comparison.

A study can use one, both, or neither.

When Are Observational Studies Needed?

A well-designed randomized experiment strengthens causal interpretation but researchers cannot always assign the conditions they want to study.

- **An experiment would be unethical.** Researchers should not assign people to smoke cigarettes to investigate long-term health outcomes.
- **The condition cannot be assigned.** Researchers studying age and memory must observe people of different ages.
- **The research concerns existing conditions.** Ecologists often compare wildlife populations across habitats that have already experienced different levels of disturbance, such as logging, pollution, wildfire, or urban development. Researchers observe and measure these existing conditions rather than deciding which habitats will be disturbed. Therefore, differences in wildlife populations may be associated with disturbance but they may also be influenced by other habitat characteristics.

Lurking Variables: What Else Might Explain Growth?

Suppose an observational study finds that fertilized seedlings grow more than unfertilized seedlings.

- The explanatory variable is **fertilizer use**.
- The response variable is **seedling growth**.

However, growers who use fertilizer might also place their seedlings in locations with more sunglight. The researchers did not record sunlight exposure.

Lurking Variable

A lurking variable is a variable omitted from the analysis that may help explain the observed relationship. Here, **sunlight exposure** is a possible lurking variable.

Confounding: Which Factor Explains the Difference?

Now suppose nearly all fertilized seedlings receive full sunlight while nearly all unfertilized seedlings grow in shade.
The groups differ in **two ways**: fertilizer use and sunlight. The data provide little opportunity to compare fertilizer treatments under the same sunlight conditions.

Confounding

Confounding occurs when the effects of different explanatory variables on the response cannot be clearly separated. Here, the effects of fertilizer and sunlight are confounded.

How the Ideas Differ

A **lurking variable** is a relevant variable that was not measured or included in the analysis. Because it remains hidden, it may help explain an observed association **Confounding** occurs when the effects of two or more explanatory variables are mixed together, making it difficult to determine which variable is responsible for changes in the response. Confounding describes a problem with separating effects, not whether a variable was recorded.

Reducing Confounding in Observational Studies

To investigate fertilizer use, researchers could make comparisons among seedlings grown under similar conditions.

- **Restriction:** include only seedlings grown in full sunlight.
- **Matching:** compare fertilized and unfertilized seedlings with similar sunlight exposure and soil conditions.
- **Stratification:** compare fertilizer groups separately within full-sun, partial-shade, and shade categories.
- **Statistical adjustment:** include measured factors, such as sunlight and watering, in a statistical model.

A Limitation

These methods can reduce confounding when suitable comparisons are available. Unmeasured or poorly measured differences can still affect the conclusions.

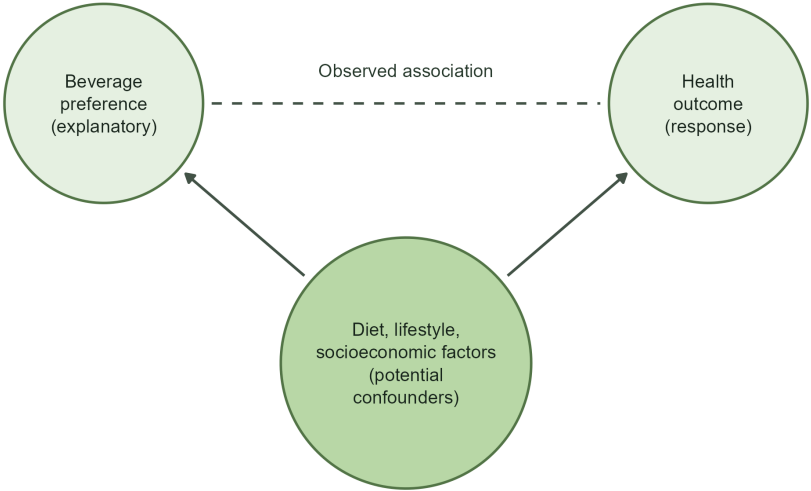
Example 6.2: Wine, Beer or Spirits?

Question

Some observational studies have reported that health outcomes differ among moderate drinkers who prefer wine, beer or spirits. Does an observed difference in health establish that beverage type itself causes the difference?

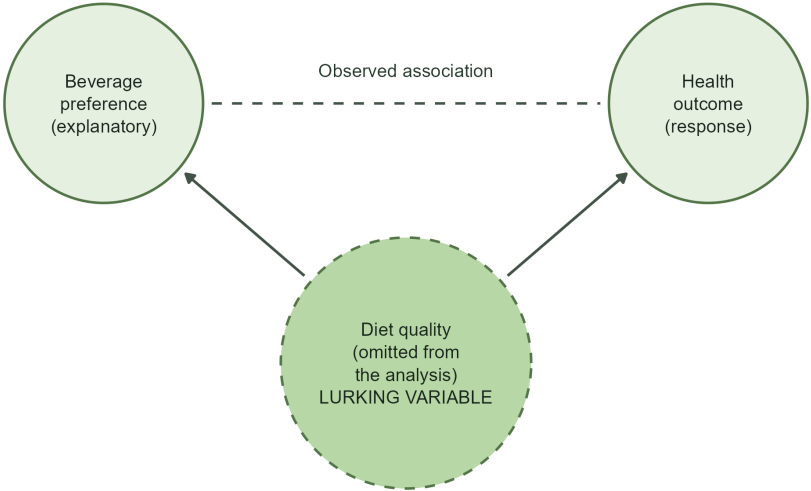
- Beverage preference can be related to many other characteristics.
- Groups that prefer different beverages may also differ in diet, smoking behaviour, exercise, income, education and access to health care.
- If these characteristics also relate to the health outcome, the effect of beverage preference cannot be separated cleanly from the effects of the other variables.

Example 6.2: Wine, Beer or Spirits?



Solid arrows show possible influences, not established causal effects.

Example 6.2: Wine, Beer or Spirits?



Possible influences are shown. The dashed border marks a variable omitted from the analysis.

Example 6.2: Wine, Beer or Spirits?

- The observational association does not by itself establish a causal effect of beverage type.
- Potential confounders provide alternative explanations for the observed difference.
- A scientifically appropriate conclusion is:
  Beverage preference and health are associated in the observed data after considering the study design. A stronger causal claim would require stronger evidence and careful control of competing explanations.

Sampling

A university health centre wants to learn about the sleep habits of first-year students. Asking every first-year student would be a **census**. Instead, the centre could collect information from **sample**: a subset of those students.

Why Sample?

A census can be expensive, slow or difficult to complete. A carefully selected sample can provide useful information with fewer resources.

The main challenge is deciding **how to select students** so that the results help us understand the larger group.

Example: Defining the Population and Variable

The health centre wants to estimate the proportion of currently enrolled first-year students who average at least 7 hours of nighttime sleep.
For this study, each student reports their average hours of night time sleep over the previous 7 nights.

Questions

1. What is the population of interest?
2. What variable determines whether a student belongs to the group being counted?

Example: Defining the Population and Variable

Population

All currently enrolled first-year students at that university. This is the group the health centre wants to describe.

Variable

Whether a student's reported average night time sleep over the previous 7 nights is at least 7 hours:

$$\text{Yes: average} \geq 7 \qquad \text{No: average} < 7$$

This is a **categorical variable**.

The reported average number of hours is a quantitative variable. We classify it into yes or no categories to answer the study's question.

From the Population to the Sample

The health centre needs a practical way to identify and select students.

Sampling Frame

The list or mechanism used to identify units that can be selected. For example, it could be the university's current first year enrolment list.

Sampling Design

The planned method for selecting units. For example, randomly select 200 distinct student IDs from that list and invite those students to complete the survey.

The **sample** provides the data. If some selected students do not respond, the health centre must consider whether respondents differ from nonrespondents.

Parameter versus Statistic

For this study, a student **meets the sleep criterion** if their reported average sleep over the previous 7 nights is at least 7 hours.

$$\text{Meets criterion: average} \geq 7 \qquad \text{Does not meet criterion: average} < 7$$

Let p be the proportion of **all currently enrolled first year students** at the university who report an average of at least 7 hours of sleep per night. This population proportion is unknown.

Statistic: Calculated from the Sample

Suppose all 200 selected students respond and 124 report an average of at least 7 hours of sleep. The sample proportion is

$$\hat{p} = \frac{124}{200} = 0.62$$

Therefore, 62% of the sampled students meet the sleep criterion.

Statistical inference uses the sample proportion $\hat{p} = 0.62$ to estimate the unknown population proportion p while accounting for sampling uncertainty.

Why the Selection Method Matters

Suppose the health centre surveys only students studying in the library at 10 p.m.

Who Might Be Missed?

Students who study elsewhere, work evening shifts, commute or go to bed earlier may be absent. Their sleep habits could differ from those of students in the library.

Selecting students randomly from a complete enrolment list gives students across these groups a chance of selection.

Sample Size Is Not Enough

Surveying more students in the same library does not remove the problem of systematically missing other students.

Planning the Sleep Survey

Before collecting data, the health centre specifies:

1. **Target population:**currently enrolled first year students at this university.
2. **Measurement:** reported average nighttime sleep over the previous 7 nights, classified using the 7-hour threshold.
3. **Sampling frame:** an updated first year enrolment list.
4. **Selection method:** randomly select 200 distinct student IDs.
5. **Missing-data plan:** arrange reminders, record response rates and specify how incomplete answers will be handled.

Generalizing from the Sample

Suppose 62% of the surveyed students meet the sleep criterion. What can the health centre conclude?

- The result estimates the proportion among **first year students at this university**.
- The design does not automatically support conclusions about students in other years or at other universities.
- Even with random selection, the estimate can vary from sample to sample.
- An incomplete enrolment list, nonresponse, or inaccurate sleep reports can introduce additional errors.

The Main Principle

A useful generalization depends on who was eligible, how students were selected and how the data were collected.

Sampling Variability versus Bias

Suppose a university repeatedly selects samples of first year students to estimate the proportion meeting a sleep criterion.

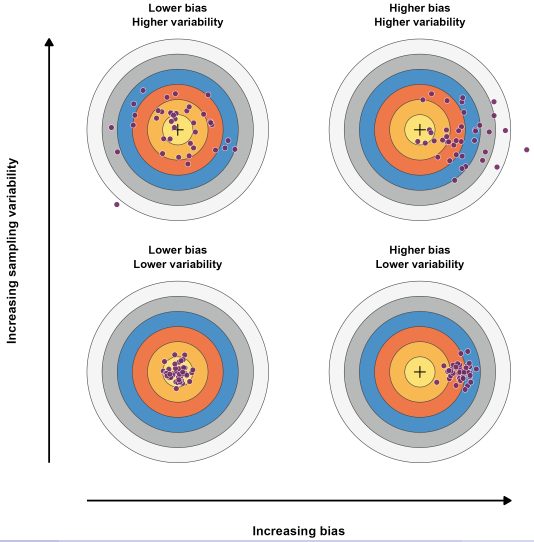
Sampling Variability: How Much Do Estimates Differ?

Different random samples usually produce different sample proportions. This natural variation from sample to sample is **sampling variability**.

Bias: Where Are the Estimates Centred?

A method is biased if its estimates systematically tend to be above or below the true population value.
For example, surveying only students in the library late at night may systematically misrepresent the university's first year students.

Archery Analogy: Bias and Sampling Variability



Reading the Four Targets

The horizontal axis compares **bias**. The vertical axis compares **sampling variability**.

- **Upper left:** estimates vary widely but are centred near the truth.
- **Upper right:** estimates vary widely and are centred away from the truth.
- **Lower left:** estimates are tightly grouped near the truth.
- **Lower right:** estimates are tightly grouped but systematically miss the truth.

Two Different Questions

How spread out are the estimates? Where is the distribution of estimates centred?

What Does Increasing Sample Size Improve?

Under the same sampling design, larger samples generally reduce sampling variability.

In the Archery Analogy

Increasing sample size can tighten the grouping of estimates. It does not necessarily move that grouping toward the bullseye.

For example, surveying more late night library users may give a more precise estimate for that group while still misrepresenting all first year students.

Precision Does Not Guarantee Accuracy

A tightly grouped set of estimates can still be systematically wrong. Good sampling requires attention to both variability and bias.

Poor Sampling Designs: Who Gets Included?

The health centre wants to estimate the proportion of first-year students who average at least 7 hours of night time sleep. Consider two ways it might collect responses:

Plan A: Ask Whoever Is Nearby

A researcher surveys students leaving a first year lecture. This is a **convenience sample**.

Plan B: Invite Anyone Interested

The centre posts an open survey link: "Are you getting enough sleep? Tell us about it!" This is a **voluntary response sample**.

In both cases, the participating students may differ systematically from the population the centre wants to describe.

Convenience Sampling: Easy to Reach

Definition

A convenience sample selects individuals or units because they are readily available.

Surveying students leaving one lecture is convenient but the sample includes only students who attend that class.

- Students in other courses have no opportunity to be included.
- Students absent from that lecture are missed.
- Attendance and course schedules may be related to sleep habits.

Why This Matters

Even a large number of responses from that lecture may misrepresent first year students across the university.

Voluntary Response: Who Chooses to Participate?

Definition

A voluntary response sample forms when an open invitation allows individuals to select themselves into the sample.

For the online sleep survey, students who are concerned about their sleep may be especially motivated to respond.

- Participation may depend on the very characteristic being studied.
- Posting the link widely does not ensure that respondents represent all first-year students.

An Important Distinction

An open survey is voluntary response sampling. Inviting randomly selected students is a different design.

Probability Samples: Selection by Chance

Veterinary researchers want to estimate the average weight of all dairy cows in a region. They cannot weigh every cow so they need a sample.

Probability Sample

A probability sample uses a random selection procedure. Every population unit has a **known nonzero probability** of being included.

For example, researchers could randomly select cow IDs from a complete regional registry.

- Researchers follow a defined selection procedure.
- Selection probabilities need not be equal.
- The design allows sampling uncertainty to be assessed.

Simple Random Sampling: Choosing Individual Cows

Suppose the registry lists 1000 dairy cows. Researchers want to select 50 distinct cows.

Simple Random Sample

In a simple random sample of size n, every possible group of n population units has the same chance of selection.

The researchers could number the cows from 1 to 1000 and use software to select 50 distinct numbers at random.

$$P(\text{a particular cow is selected}) = \frac{50}{1000} = 0.05$$

Every cow has the same inclusion probability and every possible group of 50 cows is equally likely. Selecting nearby cows until 50 are reached would not satisfy this definition.

Other Probability Designs: Sampling Across Farms

Selecting individual cows across an entire region may require considerable travel. Other designs organize selection differently.

- **Stratified sampling:** divide the registry into breed groups and select a random sample of cows from *every* breed group.
- **Cluster sampling:**randomly select farms and weigh *every* dairy cow on the selected farms.
- **Multistage sampling:** randomly select farms, then randomly select cows *within* those farms.

The Main Difference

Stratification samples within every group. Cluster sampling selects some groups. Multistage sampling adds further selection within chosen groups.

Random Sampling versus Random Assignment

Suppose the researchers also want to compare two feed supplements.

Random Sampling: Which Cows Enter the Study?

Randomly select cows from the regional registry.
This supports generalization from the sampled cows to the population covered by that registry.

Random Assignment: Which Supplement Does Each Receive?

Randomly assign the participating cows to supplement A or supplement B, then compare their weight gain.
This makes the treatment groups comparable on average and supports a causal comparison.

A study can use both procedures. Random assignment alone does not make the participating cows representative of the region.

Break Time



Sample Surveys: From Selection to Answers

Undercoverage

A **sample survey** is a study that gathers information from a sample in order to describe a population.

Example: A regional health agency wants to estimate the proportion of adults who have difficulty accessing health care. Instead of contacting every adult, it conducts a sample survey.

Three Places Where Problems Can Arise

1. **Coverage:** Who can be selected?
2. **Participation:** Who actually responds?
3. **Measurement:** How accurately do the answers describe what the agency wants to measure?

Undercoverage: Who Cannot Be Selected?

Suppose the health agency randomly selects telephone numbers from a list containing **only landlines**.

Question

Can this sample reliably represent all adults in the region, including those who use only cell phones?

Undercoverage

Undercoverage occurs when the sampling frame leaves out part of the target population. Cell-phone-only adults cannot be selected from this list.

If these excluded adults differ in access to health care, the resulting estimate can be systematically misleading.

Nonresponse versus Response Bias

Even among adults who can be selected, further problems may occur.

Nonresponse Bias: Missing Answers

Selected adults do not participate and their characteristics differ from those of respondents. For example, adults working long shifts may be harder to contact. If their access to health care also differs, the estimate can be biased.

Response Bias: Inaccurate Answers

Participants give answers that systematically misrepresent the characteristic being measured. For example, patients interviewed by their own clinic's staff may feel uncomfortable reporting dissatisfaction with that clinic.

Nonresponse concerns whose answers are missing. Response bias concerns the accuracy of the answers received.

Question Order: What Is on the Respondent's Mind?

Consider a survey that asks these questions in this order:

1. How satisfied are you with your recent dating experiences?
2. How happy are you with your life overall?



Question Order: What Is on the Respondent's Mind?

Question

Why might the first question influence the answer to the second?

The first question draws attention to dating experiences. Respondents may give those experiences more weight when evaluating their overall happiness.

Improving the Design

Researchers could randomly assign respondents to different question orders and compare the answers. This helps reveal whether order influences the results.

Survey Quality

Before trusting a survey result, ask:

1. **Population:** Who is the survey intended to describe?
2. **Coverage and selection:** Who appears in the sampling frame and how were participants selected?
3. **Nonresponse:** How many selected individuals responded and might nonrespondents differ from respondents?
4. **Question wording:** Are questions clear and neutral? For example, "How satisfied are you?" is more neutral than "How excellent was the service?"
5. **Question order and setting:** Could earlier questions or the interviewer influence the answers?

Note: A high response rate is helpful, but it does not by itself guarantee an unbiased survey.

Cohort and Case-Control Studies

Exposure

An exposure is a characteristic or experience being investigated, such as medication use or contact with a pollutant.

Cohort Study

Researchers identify a group of individuals, record their exposures and determine which individuals develop the outcome. They compare outcome rates across exposure groups.

Case-Control Study

Researchers select individuals according to whether they have the outcome, then compare their previous exposures.

Both are **observational studies**: researchers record exposures and outcomes without assigning treatments. The key distinction is how participants are selected, not simply whether the study uses historical records.

Case-Control Studies

Cases and Controls

Cases have the outcome being studied.
Controls do not have that outcome and should represent the population from which the cases arose.

Researchers compare previous exposure histories to ask: was the exposure more common among cases than among controls?

- The design is efficient for **rare outcomes** because researchers deliberately recruit cases.
- Controls need not be completely healthy: they must be suitable comparisons for the outcome studied.
- Differences in exposure histories may reflect an association, but confounding and recall bias must be considered.

Case Study: Reye's Syndrome and Aspirin

Reye's syndrome is a rare illness affecting the brain and liver, mainly in children and teenagers following a viral illness. Researchers investigated whether aspirin use during the earlier illness was associated with developing the syndrome.

- **Cases:** children diagnosed with Reye's syndrome.
- **Controls:** children without the syndrome who had experienced a similar preceding illness.
- **Exposure:** aspirin use before the syndrome developed.

Questions

1. Why is case-control sampling efficient here?
2. Can the proportion of cases in the study tell usow common the syndrome is in the population?

Case Study: What Can This Design Tell Us?

Why Is It Efficient?

Researchers begin with identified cases rather than following a very large group while waiting for a rare outcome to occur. Controls provide a comparison for previous aspirin exposure.

Why Does the Sample Not Give Disease Frequency?

Researchers choose how many cases and controls to recruit.
For example, recruiting equal numbers would make 50% of the sample cases. This would not mean that 50% of the population has the syndrome.

Appropriately designed case-control studies can estimate exposure–outcome associations, commonly using an **odds ratio**. The chosen case-control ratio alone cannot estimate population prevalence or incidence.

Cohort Studies: From Exposure to Outcome

Cohort

A cohort is a defined group of individuals whose outcomes are studied over time.

Prospective Cohort Study

Researchers record exposures before the outcomes of interest occur then follow participants to identify new outcomes.
They compare outcome rates across exposure groups without assigning the exposures.

- Following individuals over time allows researchers to estimate **incidence**: the occurrence of new cases
- The same cohort can be studied for several outcomes.
- Long follow-up can require substantial time and resources.

Case Study: The British Doctors Study

Beginning in 1951, researchers collected information about British doctors' smoking habits and subsequently recorded deaths and their causes. The 50 year report followed 34439 male doctors through 2001. Smoking information was updated periodically.

- **Exposure:** smoking history, including whether participants continued smoking or stopped.
- **Outcomes:** death from any cause and deaths from particular diseases.
- Researchers observed smoking behaviour; they did not assign doctors to smoke.

Questions

1. What does following these doctors over time allow researchers to estimate?
2. Why does this remain an observational study?

Case Study: The British Doctors Study

1. What Can Researchers Estimate?

Following the doctors over time allows researchers to estimate **death rates** and compare them across smoking groups.
They can examine deaths from any cause as well as deaths from particular diseases. These comparisons help assess the association between smoking history and later mortality.

2. Why Is This an Observational Study?

The researchers recorded the doctors' smoking behaviour; they did not assign participants to smoke, stop smoking, or remain nonsmokers.
Following people over time does not make a study an experiment. An experiment requires researchers to impose treatments.

Because smoking groups were not randomly assigned, other differences between the groups may contribute to differences in death rates.

Case Study: The British Doctors Study

Limitations to Consider:

- **Confounding:** smoking groups may differ in age, alcohol use, or other relevant characteristics.
- **Loss to follow-up:** missing outcomes can bias comparisons if those lost differ systematically.
- **Generalization:** findings from male doctors do not automatically describe every population.

Comparing Observational Study Designs

Feature	Cross-sectional survey	Case-control study	Cohort study
Starting point	Sample from a population	Cases and controls selected by outcome	Defined group with exposure information
What is compared?	Characteristics measured at one time	Previous exposures of cases and controls	New outcomes across exposure groups
Main strength	Describing a population at a given time	Studying rare outcomes efficiently	Measuring new outcomes over time
Estimates prevalence?	Yes, with suitable sampling	Not from the case-control ratio	Possible, if the required data are collected
Estimates incidence?	Not from a single snapshot	Not from the case-control ratio alone	Yes, with suitable follow-up
Main concerns	Undercoverage, nonresponse, inaccurate answers	Control selection, recall bias, confounding	Confounding, loss to follow-up, time and cost

These categories can overlap: surveys may collect data within cohort or case-control studies.

From Observational Studies to Experiments

Suppose researchers observe that fertilized seedlings grow more than unfertilized seedlings.

Observational Study: What Else Might Explain the Difference?

The groups may also differ in sunlight, soil, or watering. Researchers must consider these alternative explanations before attributing the difference to fertilizer.

Randomized Experiment: Create Comparable Groups

Researchers randomly assign seedlings to receive fertilizer or water alone and keep other growing conditions comparable.
Random assignment helps balance other characteristics between the groups, strengthening the causal comparison.